Fake News Detection- Assignment Report

Problem Statement

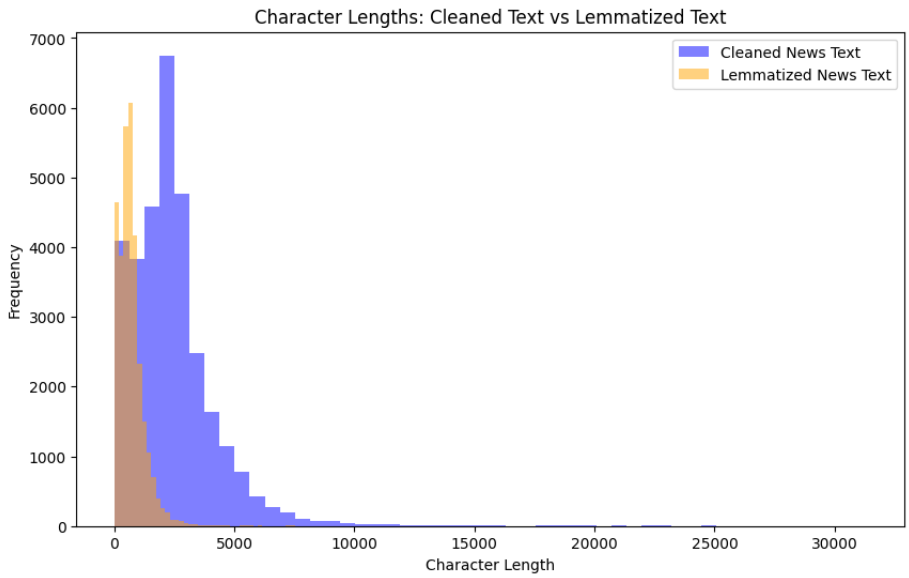
The goal of this assignment is to build an NLP classification model that can read a news article and identify whether it is "True" or "Fake".

Methodology & Techniques Used

- **Data Preparation:** Combined True.csv and Fake.csv, created a news_label (1 for True, 0 for Fake), and merged the Title and Text columns to get all the info in one place.
- **Text Cleaning:** Converted everything to lowercase, and stripped out punctuation, numbers, and bracketed text.
- **Text Preprocessing (spaCy):** Used POS tagging and lemmatization. I filtered out standard stopwords and kept only the Nouns. This helped focus strictly on the core subjects of the articles.
- **Feature Extraction (Word2Vec):** Used the pre-trained word2vec-google-news-300 model to convert the cleaned text into numerical vectors. It seems to be a well-known popular model so, used it. This captures the actual semantic meaning of the words instead of just counting them.

Exploratory Data Analysis (EDA)

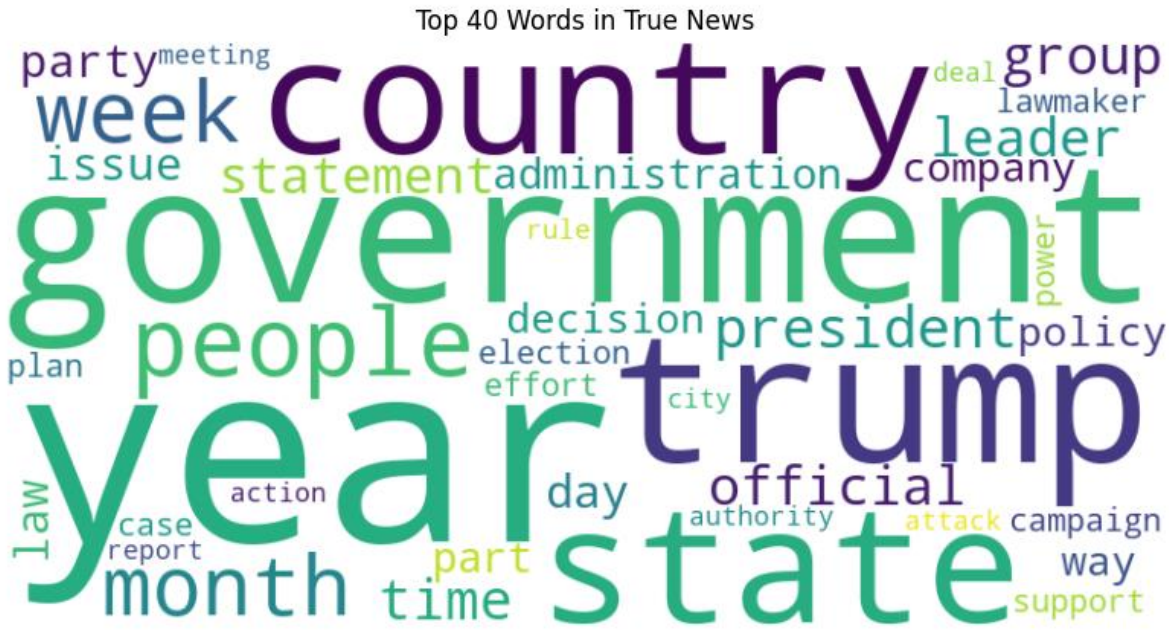
Character Length Analysis



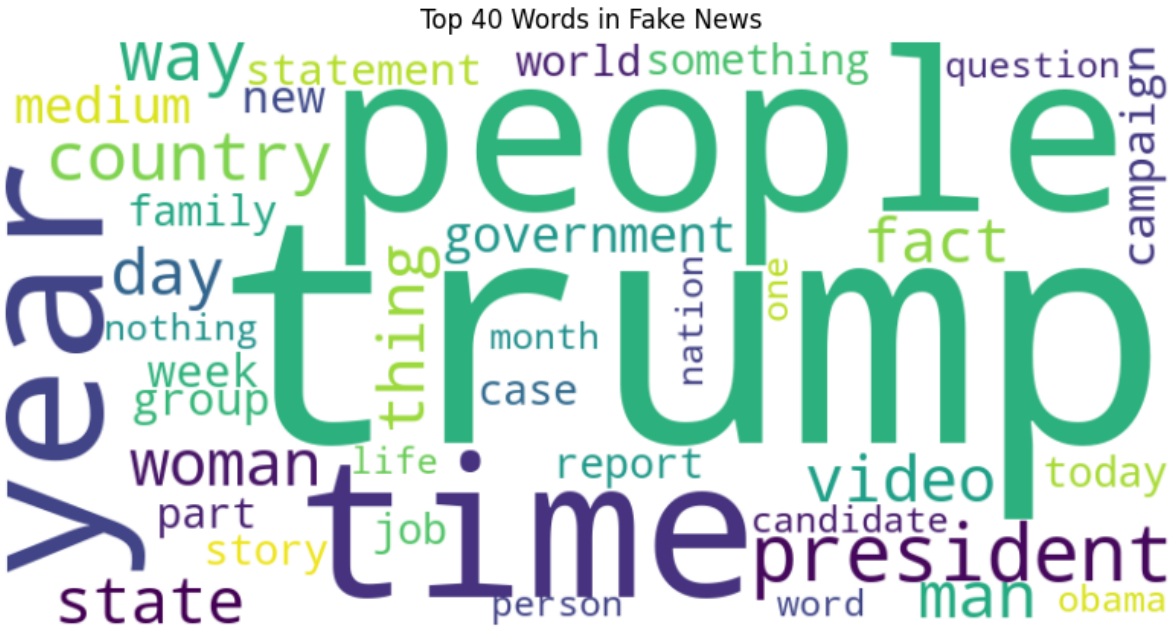
The lemmatized text is much shorter and narrower compared to the original cleaned text. This makes sense because we threw away all verbs, adjectives, and stopwords, stripping the text down to its bare bones.

Word Clouds

True:



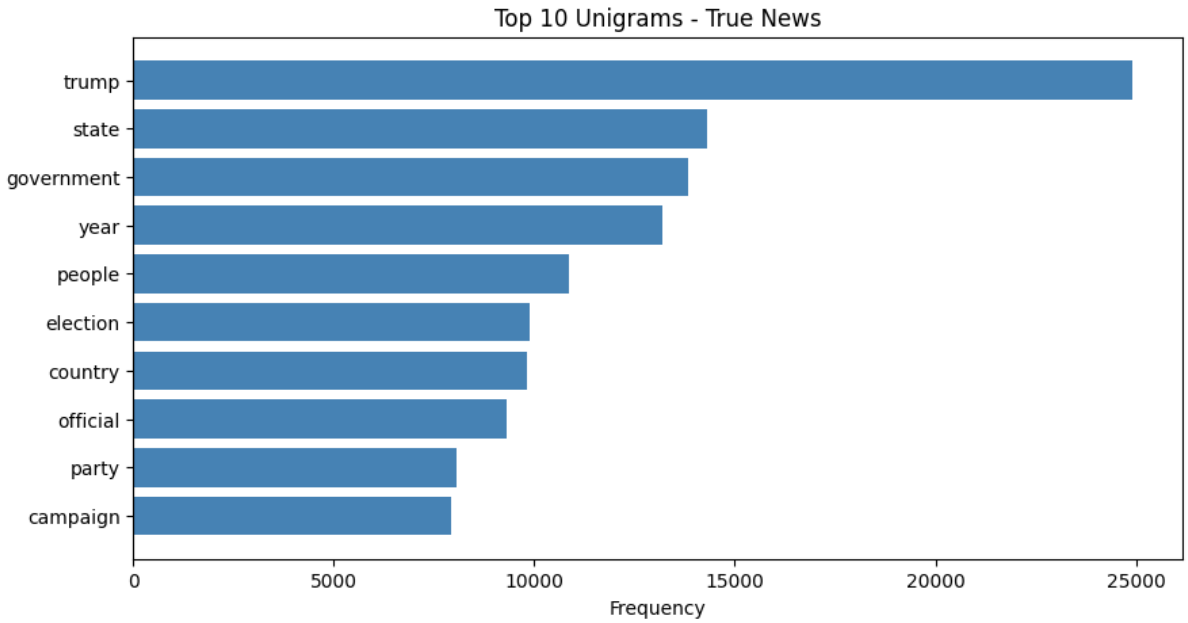
Fake:

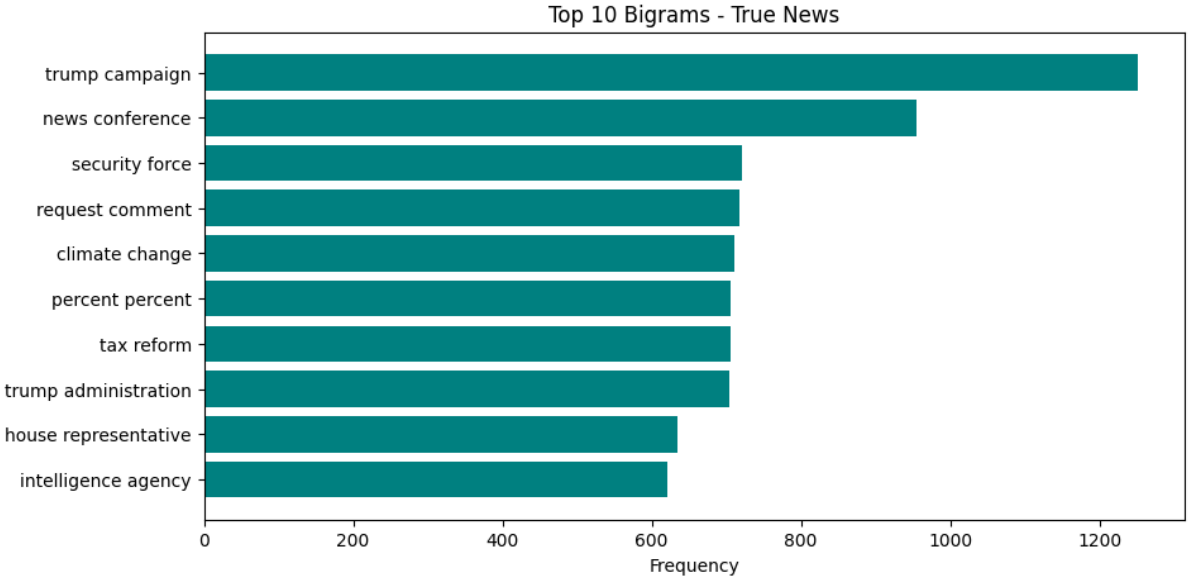


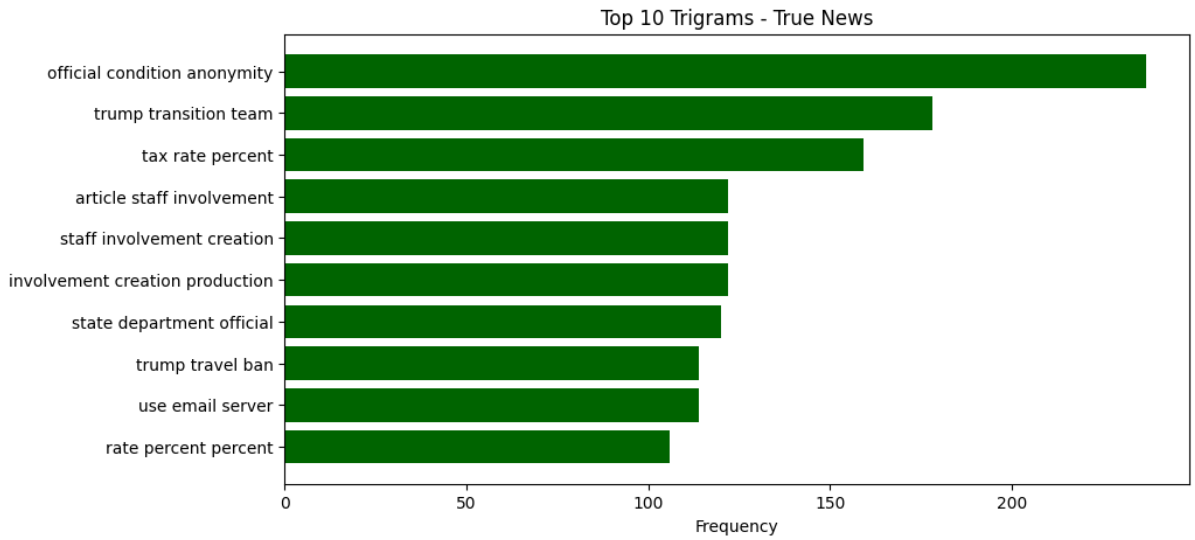
Both real and fake datasets heavily feature words like Trump, state and government showing that political topics dominate both categories, making it tricky to classify without proper NLP.

N-Grams

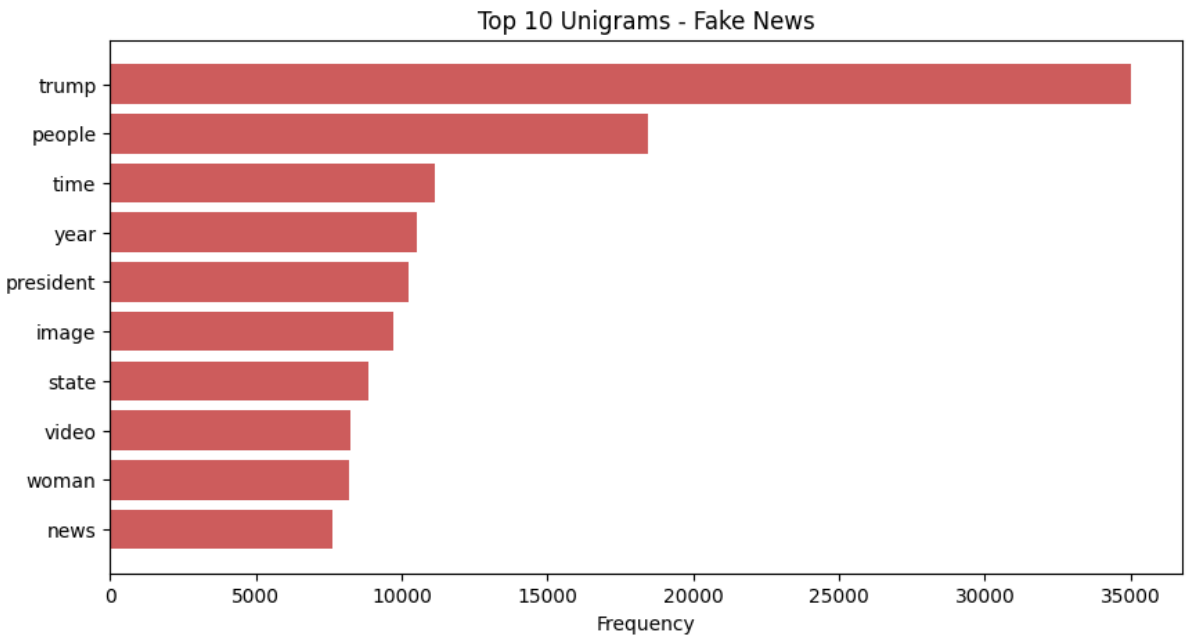
True

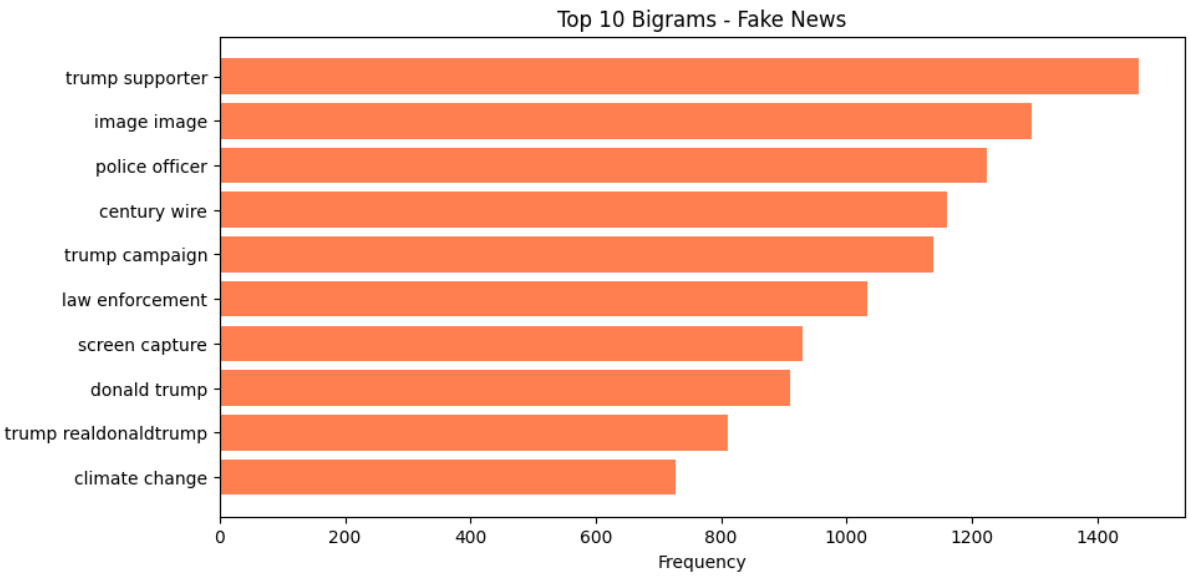

Top 10 Unigrams - True News
trump
state
government
year
people
election
country
official
party
campaign
0
5000
10000
15000
20000
25000
Frequency

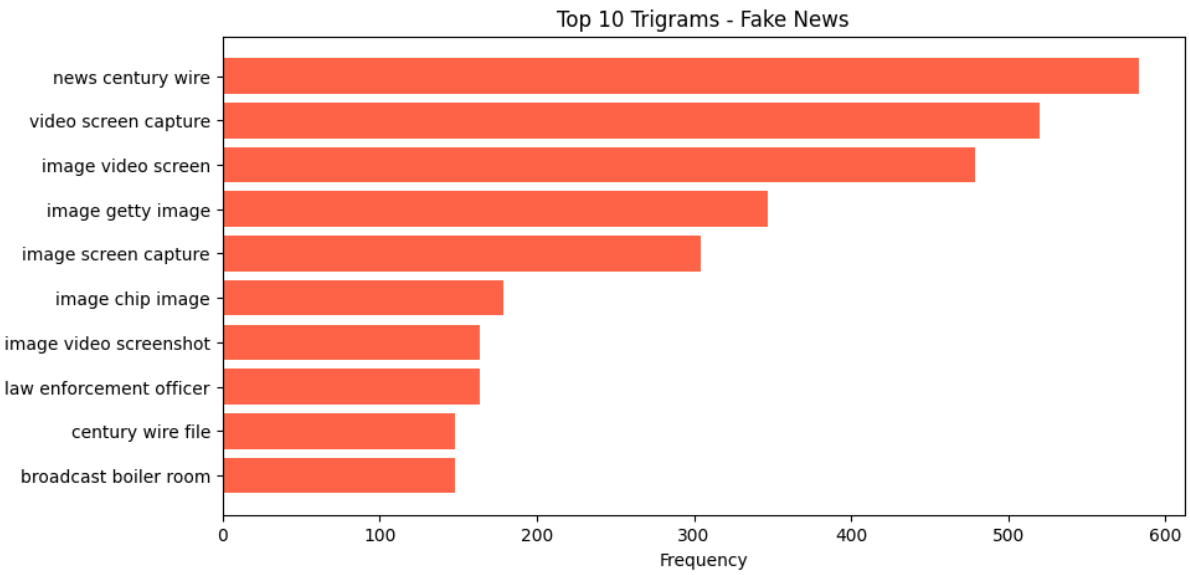

Top 10 Bigrams - True News
trump campaign
news conference
security force
request comment
climate change
percent percent
tax reform
trump administration
house representative
intelligence agency
0
200
400
600
800
1000
1200
Frequency


Top 10 Trigrams - True News
official condition anonymity
trump transition team
tax rate percent
article staff involvement
staff involvement creation
involvement creation production
state department official
trump travel ban
use email server
rate percent percent
0
50
100
150
200
Frequency

Fake


Top 10 Unigrams - Fake News
trump
people
time
year
president
image
state
video
woman
news
0
5000
10000
15000
20000
25000
30000
35000
Frequency





Looking at Unigrams (single words) isn't always helpful on its own. Bigrams and Trigrams gave much better, understandable context about what the articles were actually discussing.

Model Performance & Final Insights

We trained three different models on our Word2Vec features. Here are the results:

- **Logistic Regression:** ~90% Accuracy
- **Random Forest:** ~90% Accuracy
- **Decision Tree:** ~82% Accuracy

Conclusion: Logistic Regression and Random Forest performed good. Logistic Regression was able to easily draw a boundary to separate the Fake news from the True news.

Decision Tree struggled a bit more because it attempts to make rigid splits, which is hard to do on continuous, abstract embeddings.